

Ayan Ambesh

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SKILLS

Web Application Pentesting
Mobile Application Pentesting
Network Penetration Testing
Active Directory Exploitation
SAST & DAST
Privilege Escalation
OWASP Top 10
OSINT

Tools

- Burp Suite
- Nmap
- Metasploit
- BloodHound
- Mimikatz
- Impacket Suite
- Frida
- Wireshark
- Nessus
- YARA
- Cellebrite UFED

Programming

- Python • Bash • Rust

Platforms

- Linux
- Windows
- Active Directory
- Docker
- Git

EDUCATION

NFSU, Delhi

MSc in Digital Forensics
& Information Security
2025 – 2027 (Ongoing)

GGSIU, Delhi

BCA
2020 – 2023

St. Xavier Sr. Sec. School

Senior Secondary
2020

COMMUNITY TALK

DEF CON Groups VR (DCGVR) Talk – Java & Smali: Bridging the Gap

youtu.be/NyuJ_IK6QsU | Aug 2024

Delivered a technical talk on structural and syntactic mapping between Java and Smali for Android reverse engineering.

SUMMARY

Offensive security professional specializing in Vulnerability Assessment and Penetration Testing (VAPT) across web applications, mobile applications, and Active Directory environments. Experienced in static and dynamic application security testing (SAST/DAST), privilege escalation, and OSINT-driven reconnaissance, with practical depth built through 100+ mobile application security assessments and extensive HackTheBox/CTF exploitation. Builder of open-source security tooling, including a custom Python-based SAST scanner for vulnerability detection at scale.

EXPERIENCE

Privado.ai, Remote

Security Researcher (Privacy)

June 2023 – August 2023

- Conducted static application security testing (SAST) and code review of 100+ Android applications to identify privacy leaks, hardcoded keys, and security vulnerabilities.
- Performed dynamic application security testing (DAST) using Frida and Burp Suite to intercept API traffic, bypass SSL pinning, and triage mobile application threats.
- Built custom Python automation scripts to parse network logs and intercept flows, accelerating security vulnerability triage.
- Presented vulnerability findings and mitigation strategies directly to stakeholders to guide remediation efforts.

Forensic Science Laboratory, Rohini, Delhi

Digital Forensics Intern

May 2026 – June 2026

- Extracted and analyzed digital evidence from mobile and DVR systems using Cellebrite UFED/Premium and Tableau under forensic expert supervision.
- Assisted Crime Branch, CBI, and ACB cases by analyzing digital media while maintaining a strict chain of custody lifecycle.
- Documented detailed case findings, evidence logs, and forensic examination reports.

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PROJECTS

SaPyScan – Python Static Application Security Testing Tool

github.com/AYAN-AMBESH/sapyscan

- Built an AST-based SAST scanner in Python that performs taint tracking through assignments, imports, and method calls to detect injection sinks, with sanitizer-aware analysis to reduce false positives.
- Implemented detection rules covering OWASP Top 10 risk classes – SQL/command injection, XSS, insecure deserialization, SSRF, path traversal, hardcoded secrets, weak crypto, and SSL misconfiguration.
- Added multiprocessing support for large-codebase scanning, SARIF output for CI/CD pipeline integration, inline suppression syntax, and an automated SQL-injection autofix mode.
- Generated interactive HTML dashboards with severity-ranked findings, code snippets, and remediation guidance to streamline triage for security reviewers.

ConcealNet (CNN-Guided Steganography)

github.com/AYAN-AMBESH/ConcealNet

- Built a red-teaming data exfiltration tool implementing CNN-guided adaptive LSB steganography to conceal data in carrier images.
- Engineered visual priority maps using CNNs and Sobel gradients, adding security through SHA-256 deterministic jitter for keyed-pixel ordering, Reed-Solomon ECC, and CRC32 checks.

Offensive Security Labs & HackTheBox

Ongoing Security Practice

- Penetrated multiple HackTheBox machines, exploiting OWASP Top 10 vulnerabilities and performing privilege escalation on Windows/Linux targets.
- Executed Active Directory exploitation paths using BloodHound for AD path analysis, Mimikatz for credential harvesting, and Impacket for lateral movement.

PUBLICATIONS

Refined Human-Computer Interaction: Enhancing Efficiency and Collaboration

link.springer.com/chapter/10.1007/978-981-97-0892-5_46

Springer | DOI:10.1007/978-981-97-0892-5_46